

HIMADRI S CHATTERJEE

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Founding Software Engineer with 4+ years of experience building and scaling production AI platforms, real-time streaming services, and payment infrastructure on GCP. Deep hands-on understanding of the transformer stack—from model training and fine-tuning to multi-agent orchestration and cloud deployment. Track record of shipping product in fast-moving startup environments, working cross-functionally with ML, product, and design teams.

TECHNICAL SKILLS

Languages: Python, Go, C, C++, SQL, JavaScript

AI / ML: PyTorch, Transformer Architectures, Multi-Agent Systems, LLM Tool-Use, Model Fine-Tuning, Gemini API

Infrastructure & Cloud: GCP (Cloud Run, GKE, IAP, Compute Engine), AWS, Kubernetes, Docker, Terraform

Backend & Data: gRPC, REST APIs, PostgreSQL, MongoDB, Firebase, Stripe Payments

Frontend & Tools: React, Next.js, HTML/CSS, Git

EXPERIENCE

Founding Software Engineer — Convai, Bangalore

Mar 2022 – Present

- Designed and maintained core REST APIs and gRPC streaming services powering real-time conversational AI interactions; achieved sub-200ms end-to-end latency for voice and text channels at scale.
- Owned early-stage infrastructure on Google Cloud Run—service orchestration, autoscaling policies, blue-green rollouts, and performance tuning across 15+ microservices.
- Built the payments platform end-to-end with Stripe: subscription billing, usage-based metering, and secure checkout flows handling thousands of monthly transactions; co-designed the broader payments architecture.
- Architected on-premise deployment strategy for enterprise clients, packaging services as containerized, GPU-enabled environments; successfully deployed to multiple enterprise accounts.
- Co-built the primary frontend application (React), implementing creator workflows, UI components, and third-party integrations used by the entire user base.
- Worked cross-functionally with ML, product, and design teams to ship features on tight timelines while maintaining platform reliability.

Systems Engineer — Tata Consultancy Services, Bangalore

May 2021 – Mar 2022

- Developed and deployed cloud-based infrastructure solutions on Azure for enterprise clients, focusing on compute provisioning, networking, and identity management.
- Built internal automation tools that streamlined operational workflows for the client organization, reducing manual effort in reporting and resource tracking.

Software Developer Intern — BNP Paribas, Mumbai

Jan 2021 – Mar 2021

- Developed full-stack features for an internal invoice management application (Angular, C#, SQL Server), improving data retrieval performance and extending UI functionality.

PROJECTS

Durable AI Agent Workflow Platform — Effectively-Once Execution Engine | [GitHub](#)

Python, FastAPI, Celery, Postgres, Redis, OpenTelemetry, Jaeger, Next.js, Docker

- Built a production-style platform that runs AI tasks as durable workflows on FastAPI, Celery, Postgres, and Redis, with Postgres as the source of truth and the queue only scheduling work.
- Hand-built the durable execution engine: commit-before-execute steps, retry with exponential backoff, idempotency keys for effectively-once side effects, and a periodic sweeper that resumes runs stranded by a worker crash.

- Added human-in-the-loop approval gates with an audit trail, OpenTelemetry distributed tracing to Jaeger, a cost/latency metrics dashboard, and an automated eval harness scoring runs across six quality and cost dimensions.
- GPT-2 from Scratch — with LoRA Fine-Tuning & Speculative Decoding | [GitHub](#)**
Python, PyTorch, BPE Tokenizer, Multi-Head Attention, LoRA, Mixed-Precision Training
- Implemented a GPT-2 (124M) language model from scratch in PyTorch—custom BPE tokenizer, multi-head self-attention, positional embeddings, and full training loop on OpenWebText with mixed-precision (bf16) and gradient accumulation.
 - Extended the base model with a LoRA adapter layer for parameter-efficient domain fine-tuning; implemented speculative decoding with a draft model to accelerate inference by ~2x while preserving output distribution.
 - Benchmarked training dynamics (loss curves, gradient norms, learning rate schedules) and evaluated perplexity and generation quality against the original OpenAI checkpoint.

- Violent Crowd Behaviour Detection — C3D Video Classification (Published) | [GitHub](#)**
Python, C3D, 3D CNN, Spatiotemporal Features, Video Classification, ViolentFlow Dataset
- Trained a C3D (3D convolutional) network on the ViolentFlow dataset to classify violent versus non-violent behaviour in crowd surveillance footage, learning spatiotemporal features directly from short video clips.
 - Built the video preprocessing pipeline—frame sampling, clip extraction, and normalization—and tuned the training setup to work with a limited set of labelled clips.
 - Evaluated classification performance on held-out footage and documented the methodology and results in a published research paper.

- Work Buddy — Workstream Control Agent (AI Agents Intensive Capstone) | [GitHub](#)**
Python, Google ADK, Multi-Agent Orchestration, MCP, Gemini / Vertex, FastAPI
- Built Work Buddy, a multi-agent system on Google’s Agent Development Kit (ADK) that turns messy project context—meeting notes, Slack, email, Jira, and repo state—into an auditable execution ledger; a coordinator delegates to five specialist sub-agents for ingestion, cross-source reconciliation, and drafting.
 - Enforced a human-in-the-loop approval gate before any external write (Jira, Slack, email, GitHub PR), with secret redaction and source-evidence tracking, so the agent never claims work is done, merged, or shipped without tool proof.
 - Exposed the full tool set over a Model Context Protocol (MCP) server, added an eval suite covering the safety-sensitive behavior, and shipped a FastAPI dashboard with a Cloud Run deployment path; model-agnostic via Gemini/Vertex with OpenAI, Claude, and Ollama fallbacks.

EDUCATION

Master of Computer Application — Vellore Institute of Technology, Vellore	2019 – 2021
BSc Computer Science — RKMRC, Kolkata	2016 – 2019

CERTIFICATIONS

Google/Kaggle AI Agents Intensive — Capstone Certified (June 2026)